

Clinical proof-of-concept

ECG anomaly detection — example beats

Model: mitbih_autoencoder_best.pt
Threshold (F1-optimal, real-time preprocessing): 0.0591
Records used: 100,101,102,103,104,105,106,107
Total beats inspected: 17355 Anomalies in set: 6890

Confusion matrix on these beats:

True positives	: 6306	(correctly caught)
False negatives	: 584	(missed by the model)
False positives	: 798	(false alarms)
True negatives	: 9667	(correctly cleared)

Sensitivity = 0.915

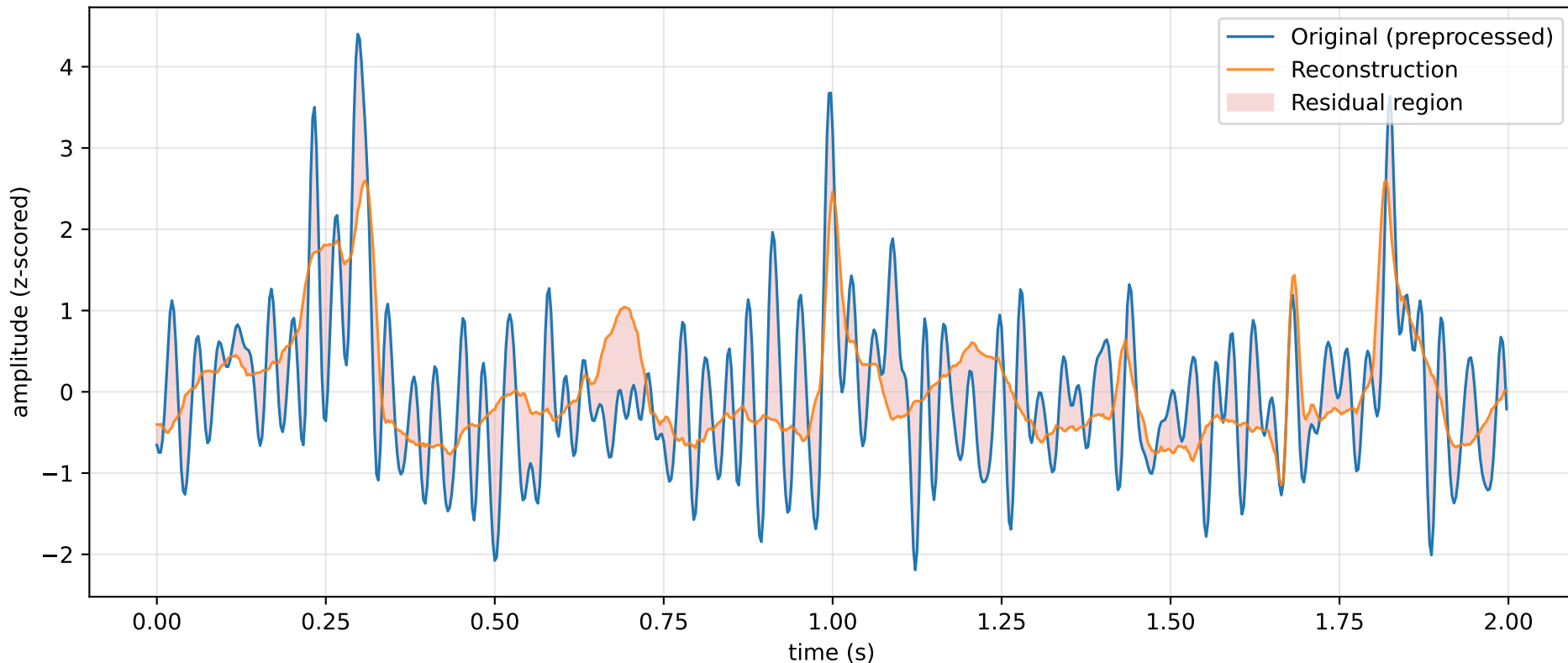
Specificity = 0.924

How to read the next pages: each page shows one 2-second window with the original ECG (blue) and the model's reconstruction (orange). The shaded red area between them is the reconstruction error. A beat is flagged when that error exceeds the threshold above.

TRUE POSITIVE — model correctly flagged abnormal

(4 example(s) follow)

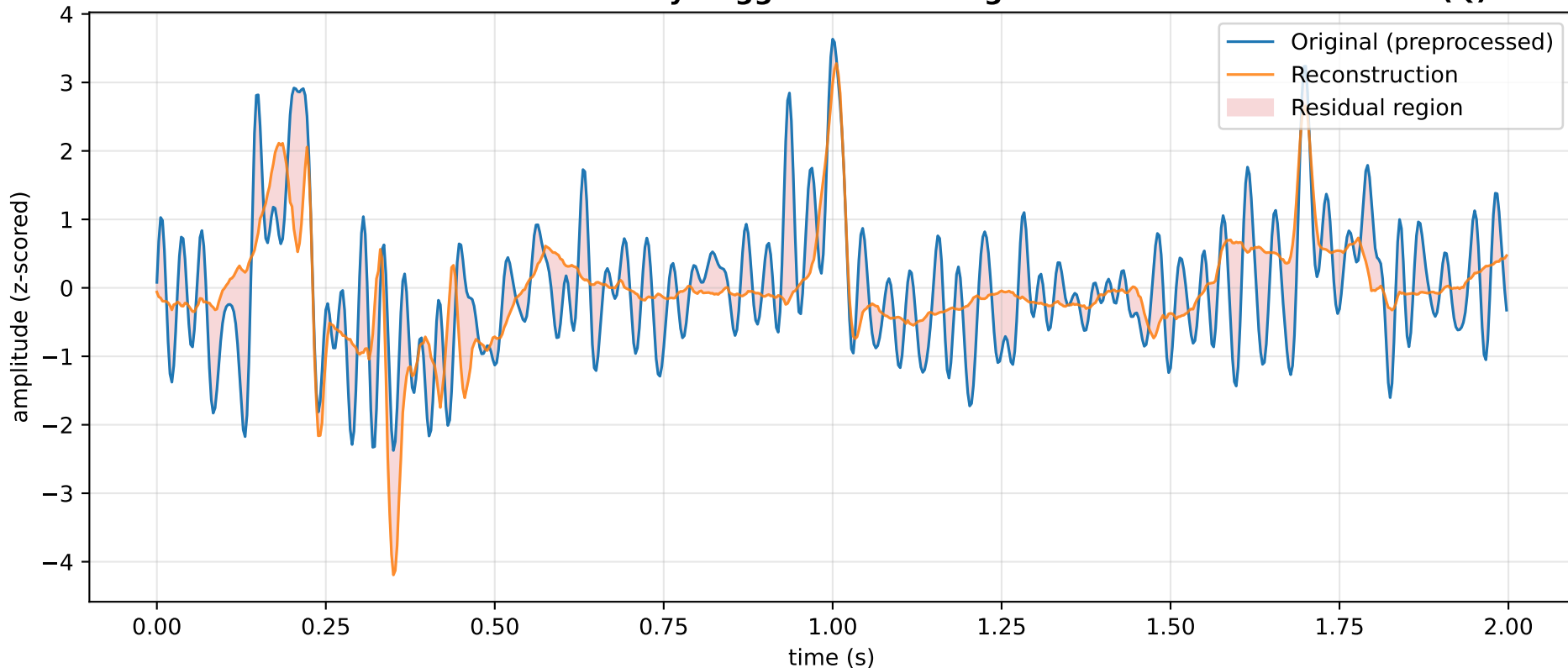
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115658
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 0.7100
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

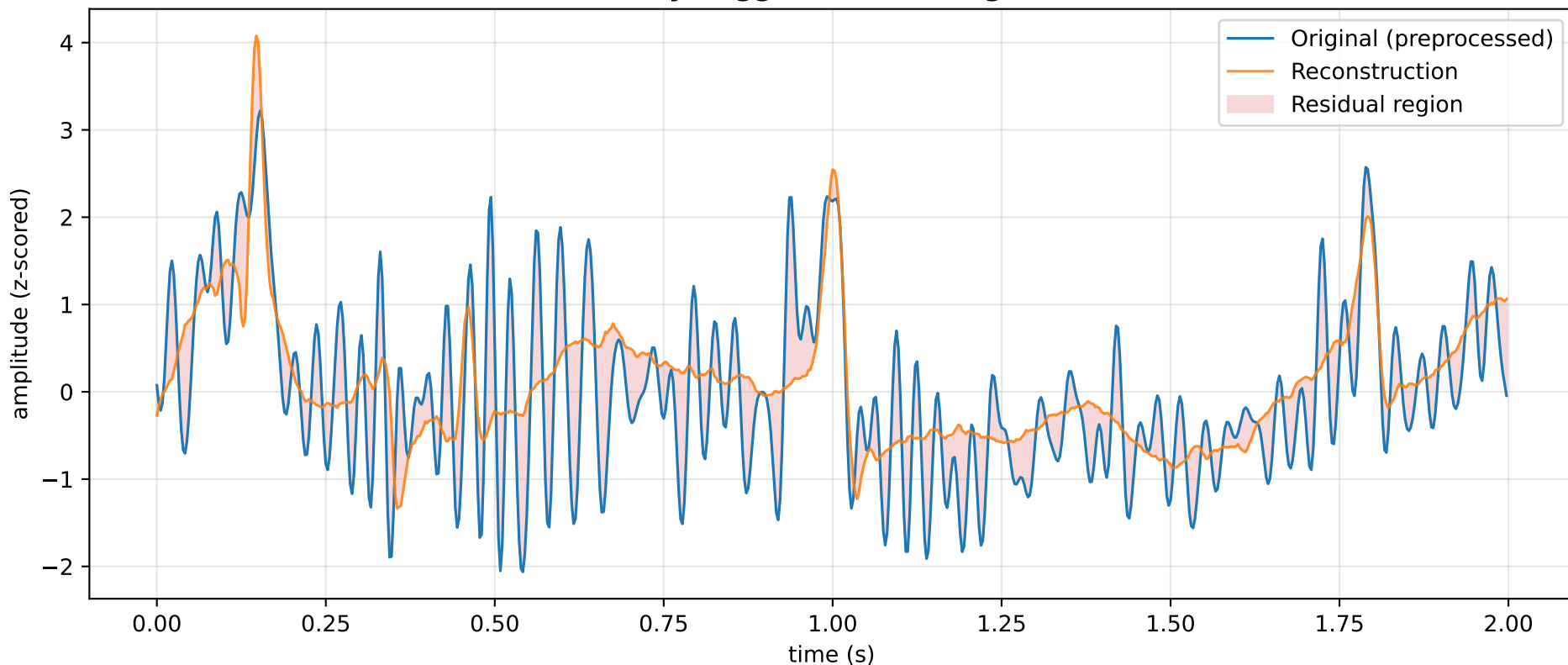
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115405
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 0.6910
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

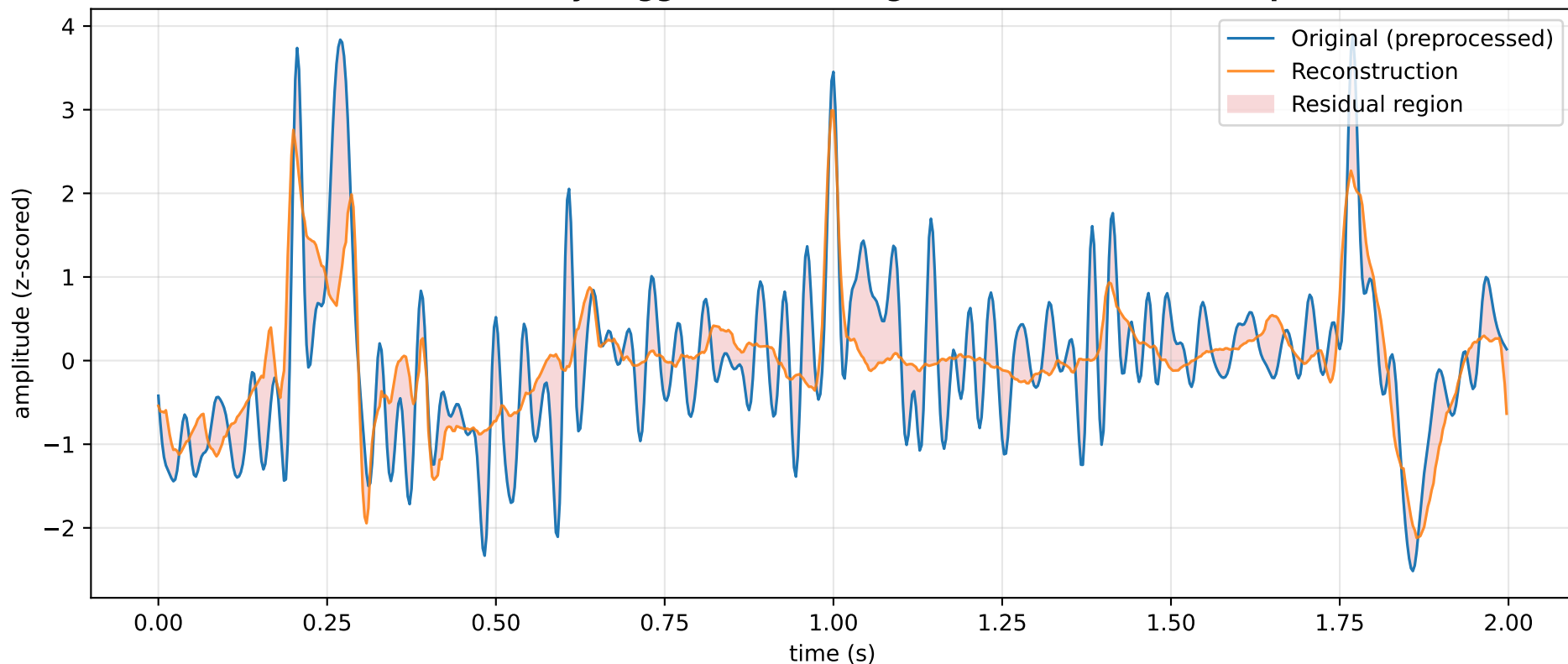
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115121
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 0.5579
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

TRUE POSITIVE — model correctly flagged abnormal: ground truth = Fusion of paced and normal (f)



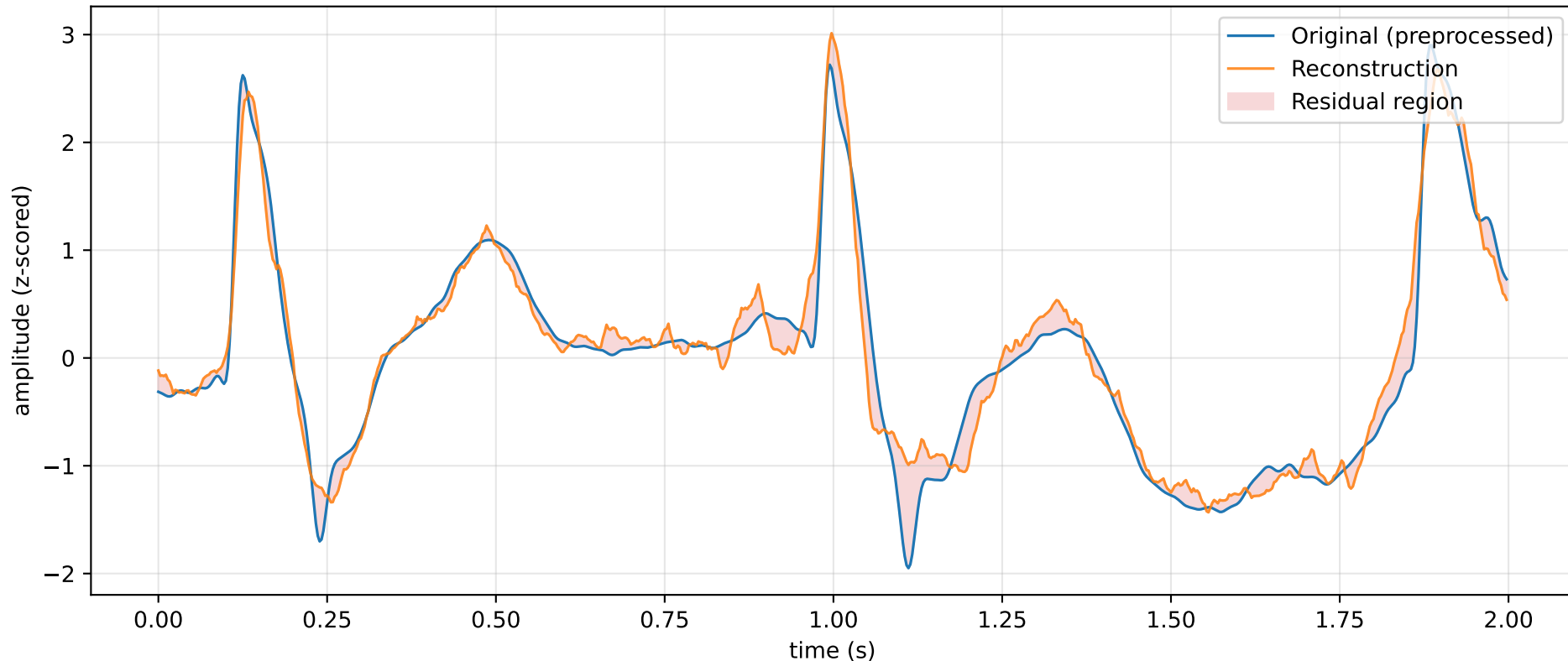
Source : rec 104, sample 50820
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.5041
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

FALSE NEGATIVE — model MISSED an abnormal beat

(4 example(s) follow)

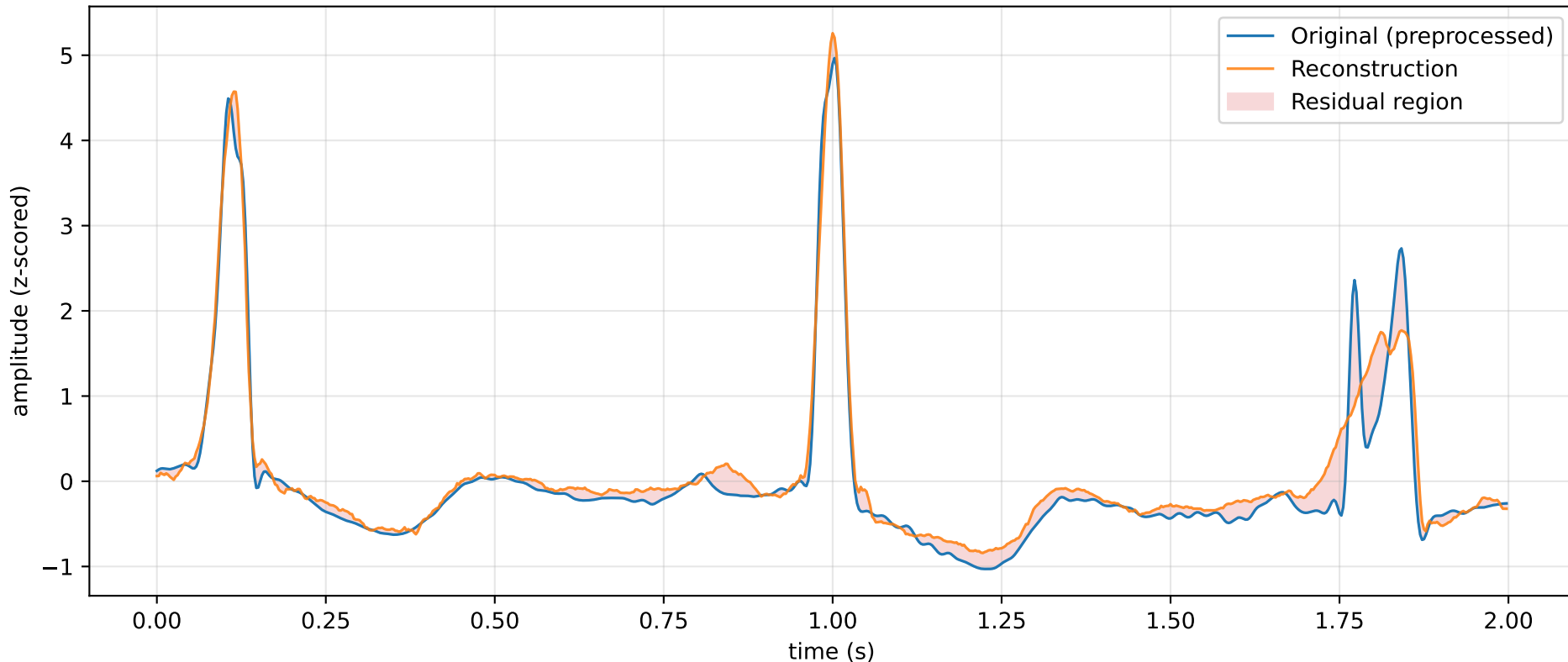
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Paced beat (/)



Source : rec 107, sample 337855
Beat symbol : / (Paced beat)
Ground truth : ANOMALY
Residual : 0.0591
Threshold : 0.0591
Model says : NORMAL

NORMAL

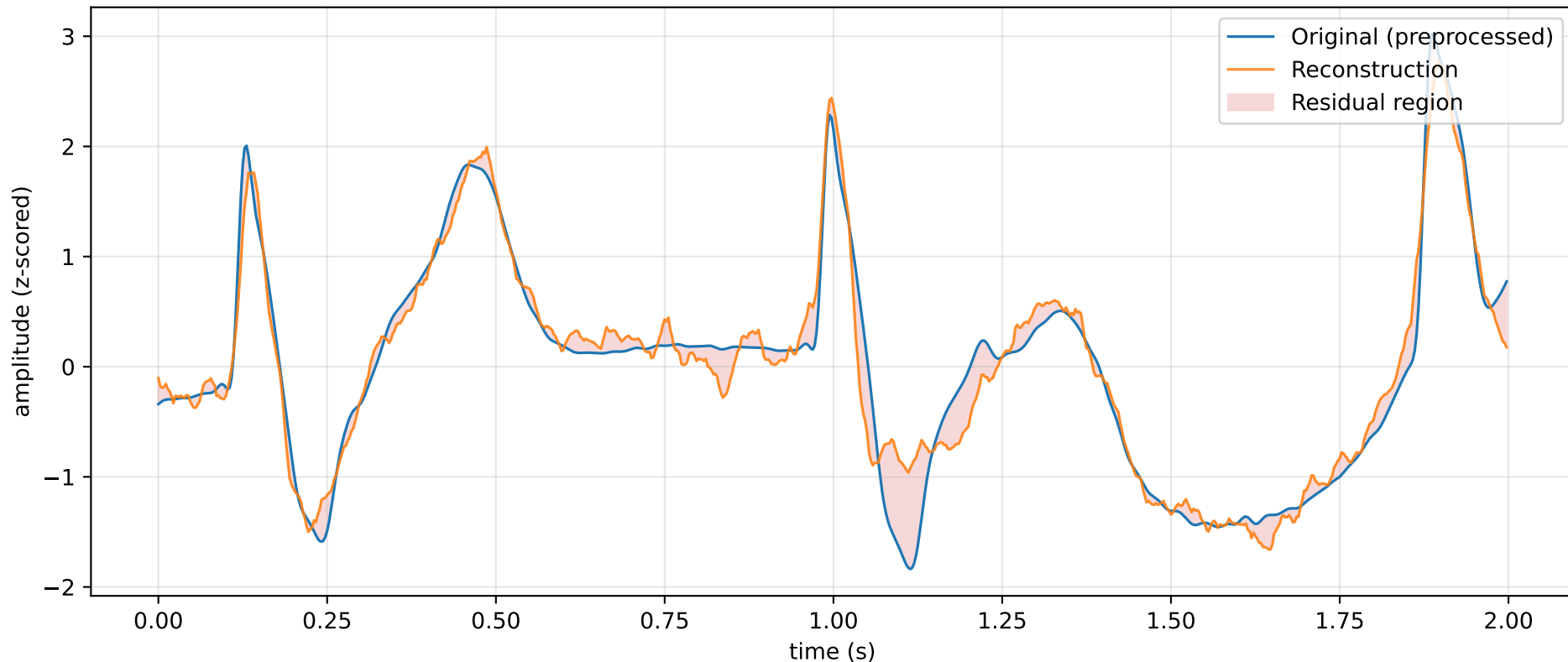
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 225360
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0591
Threshold : 0.0591
Model says : NORMAL

NORMAL

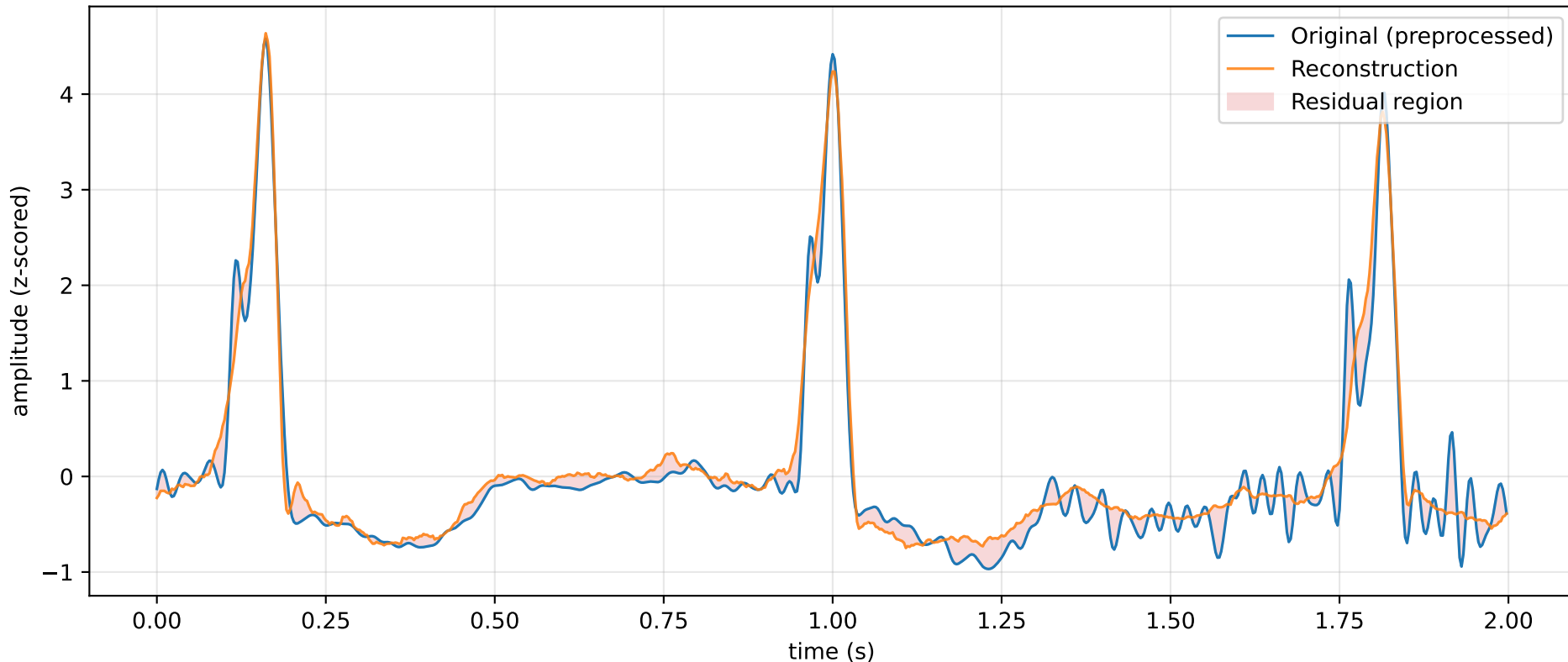
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Paced beat (/)



Source : rec 107, sample 471816
Beat symbol : / (Paced beat)
Ground truth : ANOMALY
Residual : 0.0590
Threshold : 0.0591
Model says : NORMAL

NORMAL

FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



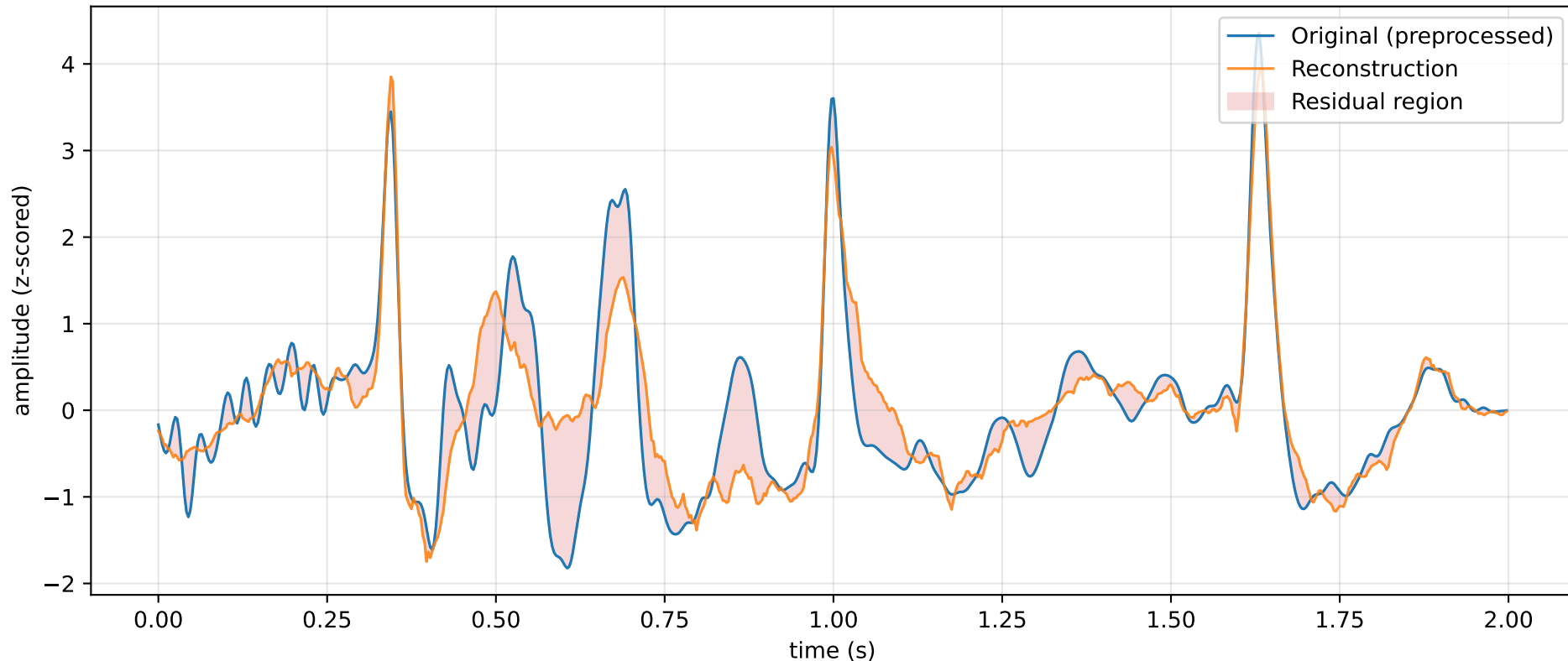
Source : rec 104, sample 136586
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0590
Threshold : 0.0591
Model says : NORMAL

NORMAL

FALSE POSITIVE — model wrongly flagged a normal beat

(4 example(s) follow)

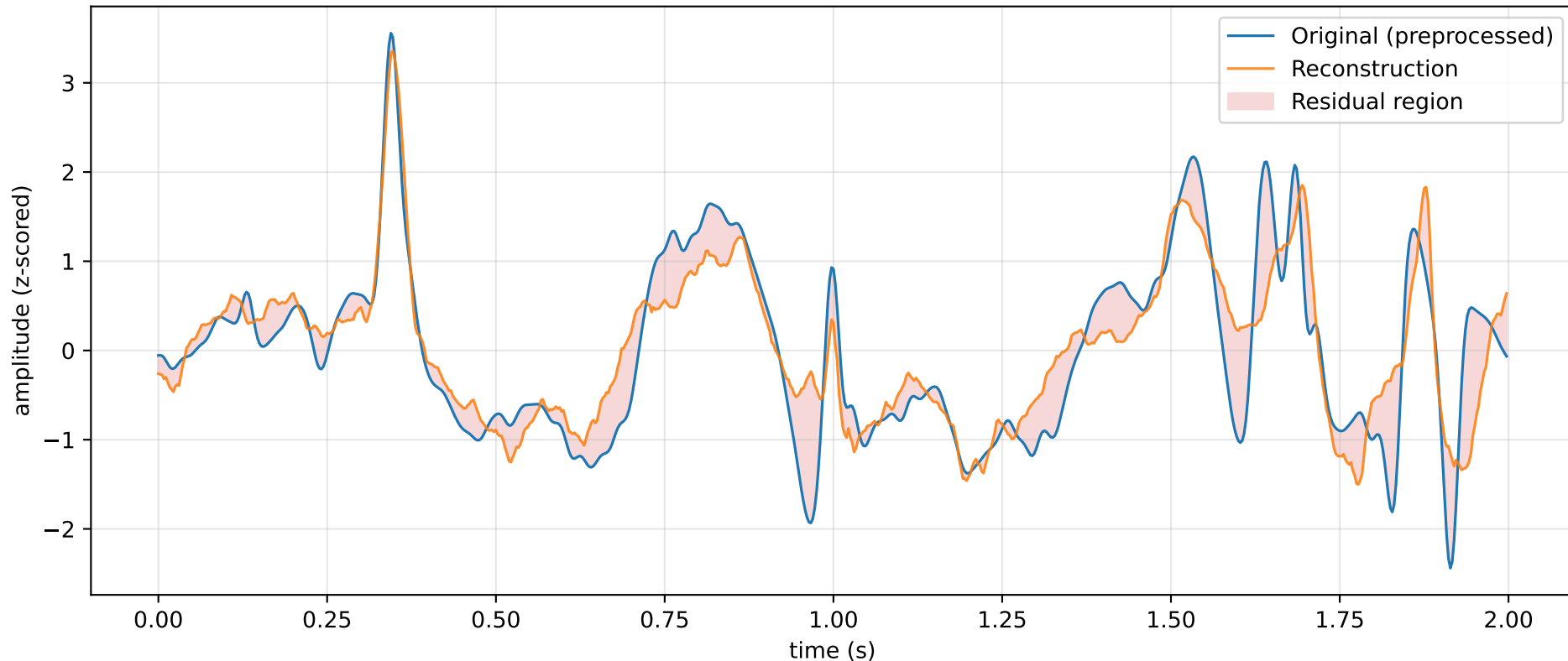
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 482302
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.2769
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

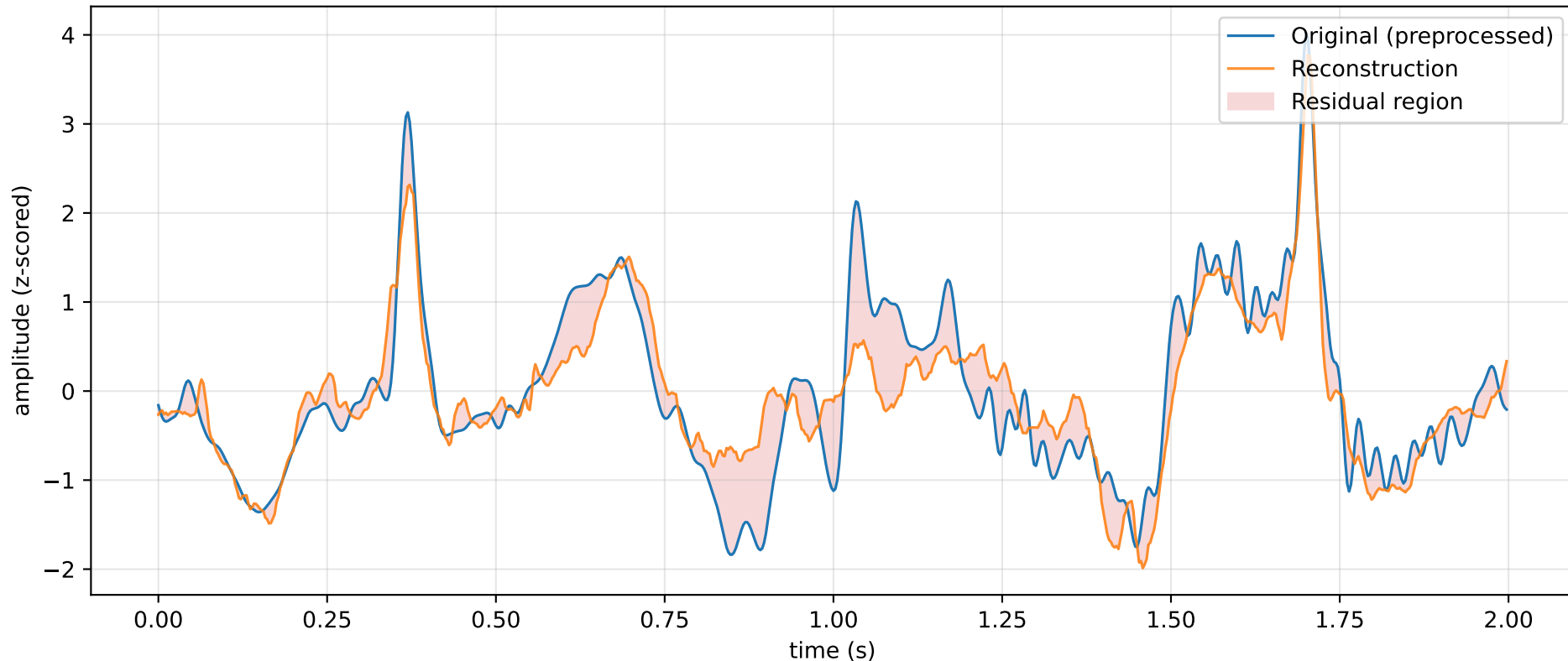
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 472351
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.2580
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

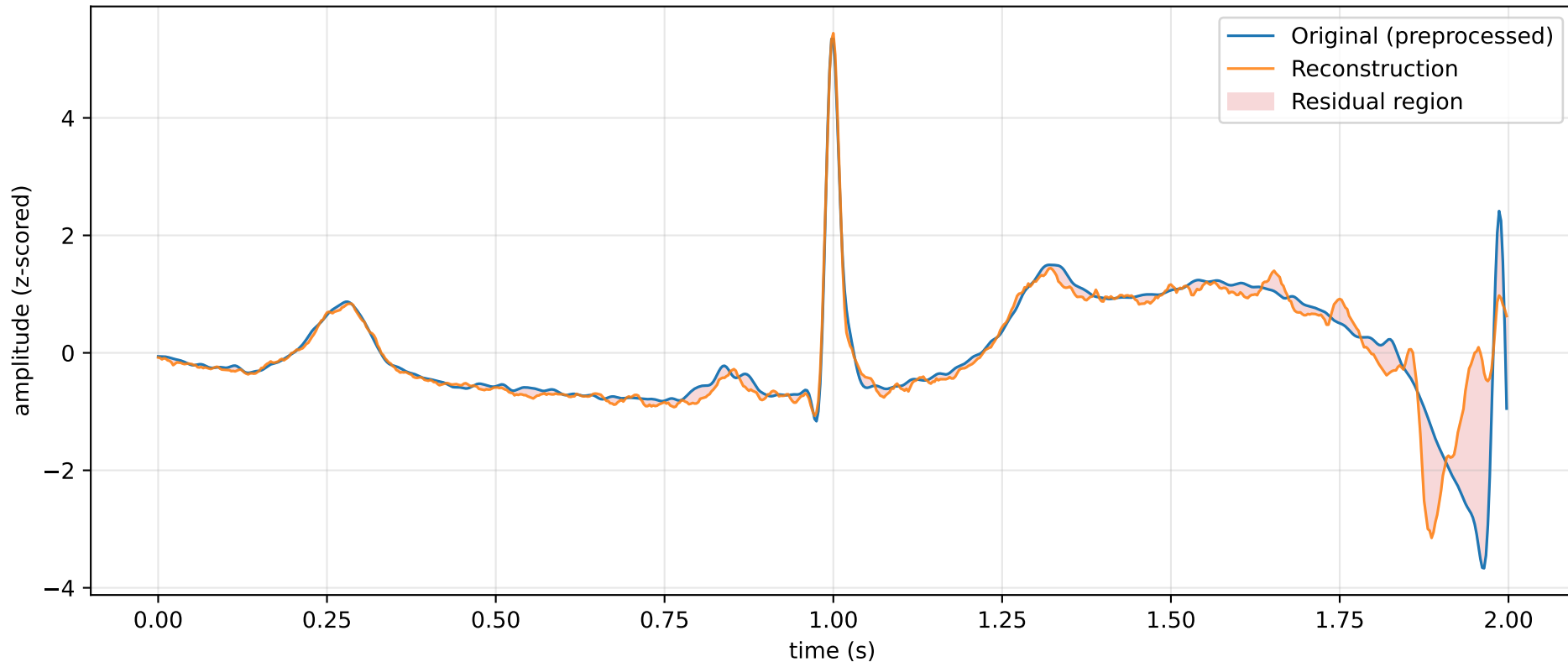
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 314406
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.2371
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



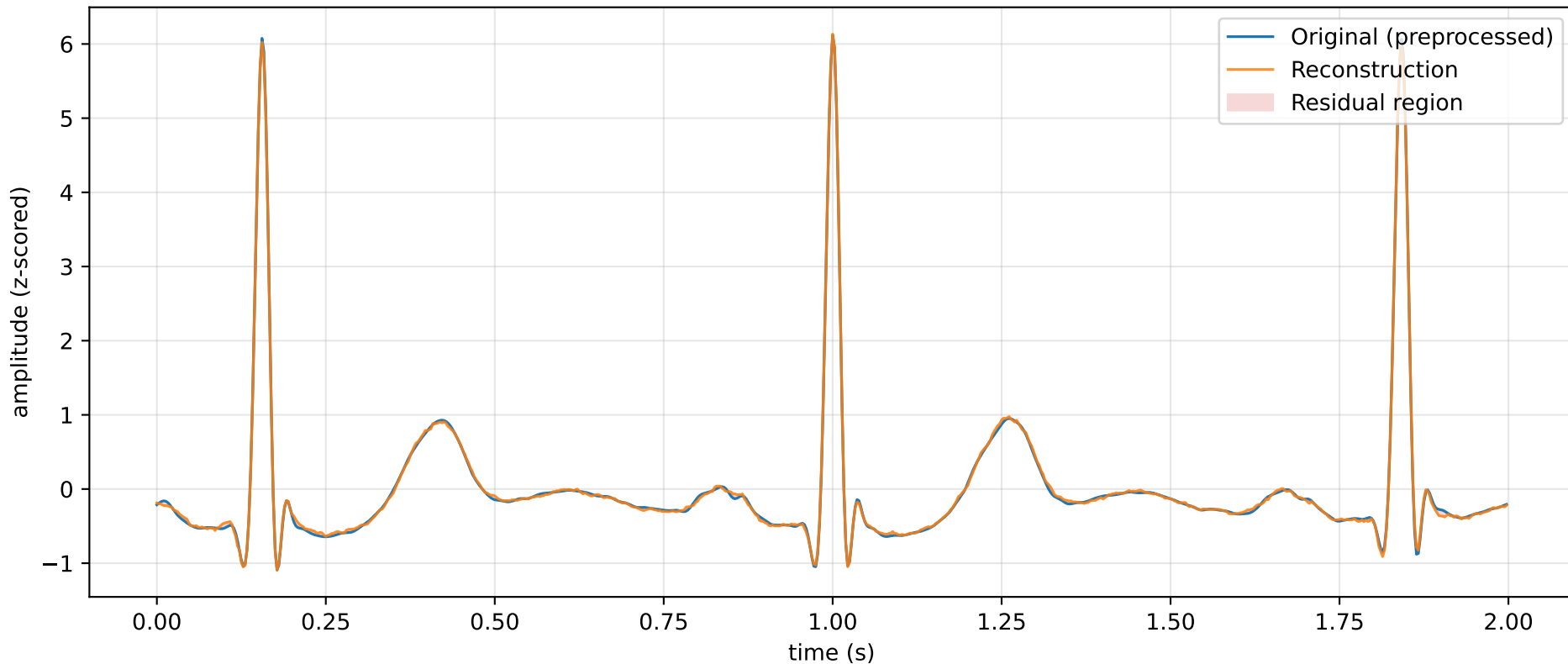
Source : rec 101, sample 463759
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.2289
Threshold : 0.0591
Model says : ANOMALY

ANOMALY

TRUE NEGATIVE — model correctly cleared a normal beat

(4 example(s) follow)

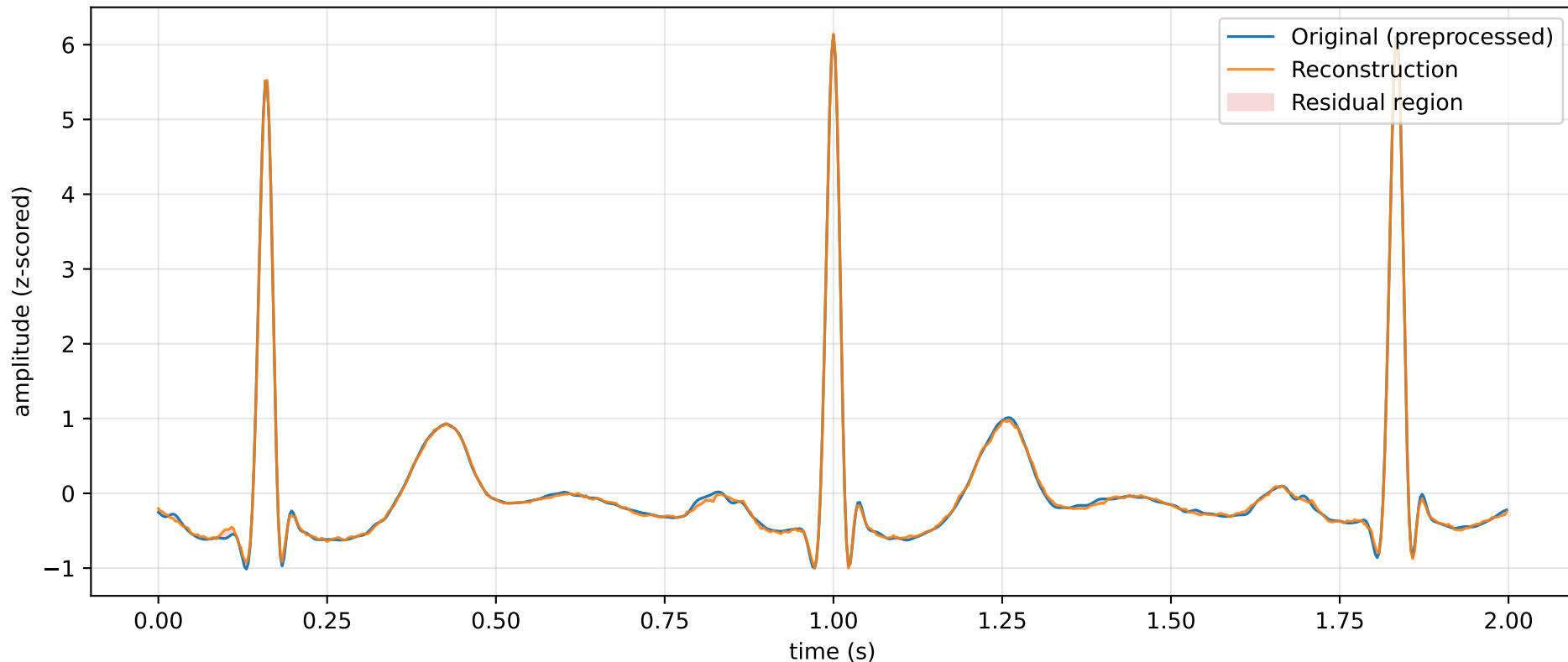
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 103, sample 263928
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0008
Threshold : 0.0591
Model says : NORMAL

NORMAL

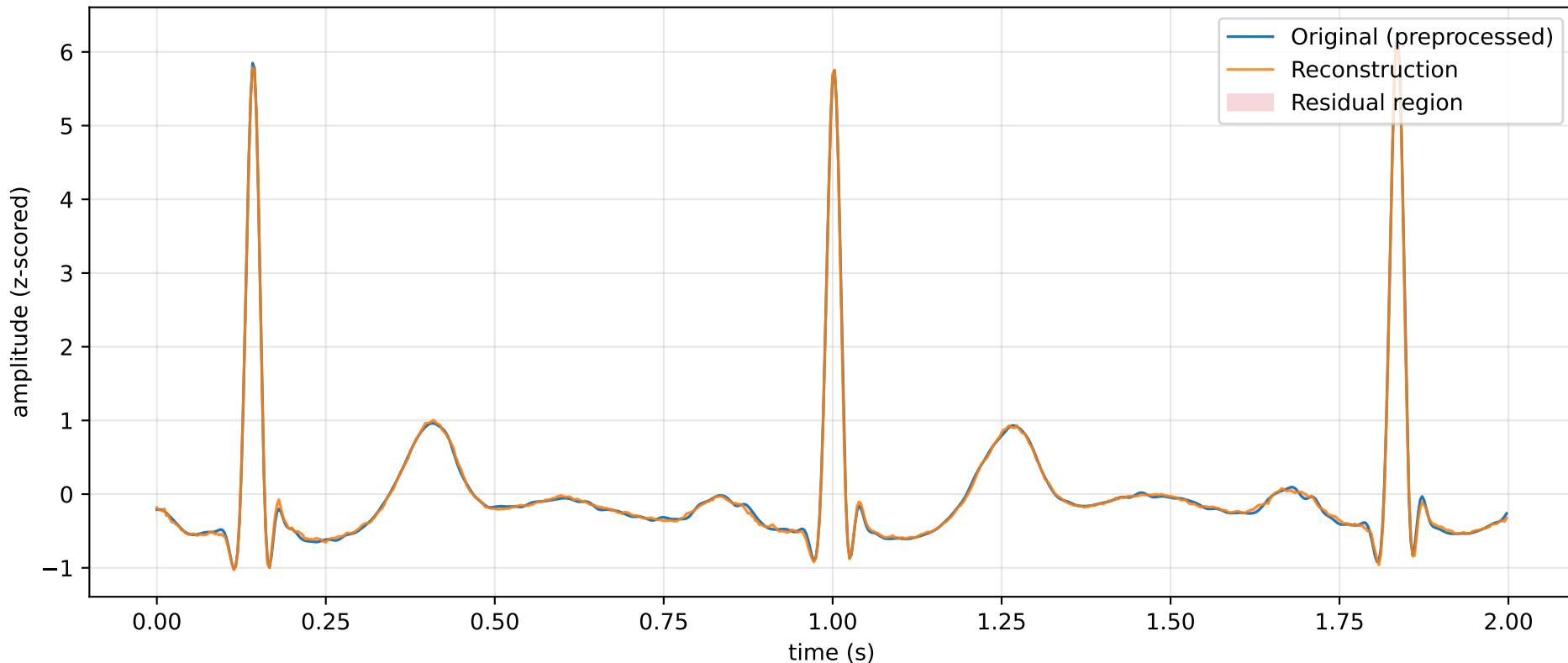
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 103, sample 37418
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0010
Threshold : 0.0591
Model says : NORMAL

NORMAL

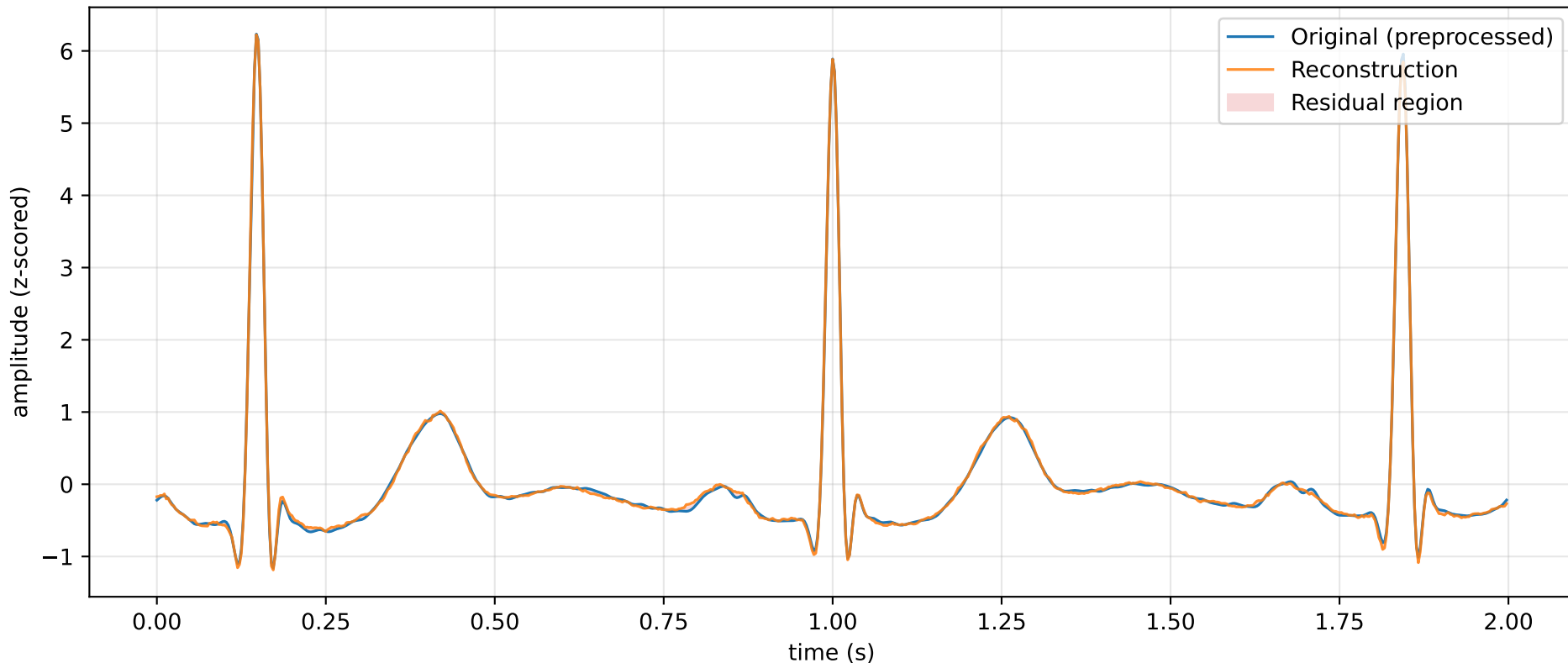
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 103, sample 34938
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0010
Threshold : 0.0591
Model says : NORMAL

NORMAL

TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 103, sample 244650
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0010
Threshold : 0.0591
Model says : NORMAL

NORMAL